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compact1, compact2, compact3

java.util.spi

Class CalendarNameProvider

java.lang.Object
    java.util.spi.LocaleServiceProvider
        java.util.spi.CalendarNameProvider

```
public abstract class CalendarNameProvider
extends LocaleServiceProvider
```

An abstract class for service providers that provide localized string representations (display names) of `Calendar` field values.

Calendar Types

Calendar types are used to specify calendar systems for which the `getDisplayName` and `getDisplayNames` methods provide calendar field value names. See `Calendar.getCalendarType()` for details.

Calendar Fields

Calendar fields are specified with the constants defined in `Calendar`. The following are calendar-common fields and their values to be supported for each calendar system.

Field	Value	Description
`Calendar.MONTH`	`Calendar.JANUARY` to `Calendar.UNDECIMBER`	Month numbering is 0-based (e.g., 0 - January, ..., 11 - December). Some calendar systems have 13 months. Month names need to be supported in both the formatting and stand-alone forms if required by the supported locales. If there's no distinction in the two forms, the

		same names should be returned in both of the forms.
`Calendar.DAY_OF_WEEK`	`Calendar.SUNDAY` to `Calendar.SATURDAY`	Day-of-week numbering is 1-based starting from Sunday (i.e., 1 - Sunday, ..., 7 - Saturday).
`Calendar.AM_PM`	`Calendar.AM` to `Calendar.PM`	0 - AM, 1 - PM

The following are calendar-specific fields and their values to be supported.

Calendar Type	Field	Value	Description
`"gregory"`	`Calendar.ERA`	0	`GregorianCalendar.BC` (BCE)
		1	`GregorianCalendar.AD` (CE)
`"buddhist"`	`Calendar.ERA`	0	BC (BCE)
		1	B.E. (Buddhist Era)
`"japanese"`	`Calendar.ERA`	0	Seireki (Before Meiji)
		1	Meiji
		2	Taisho
		3	Showa
		4	Heisei
	`Calendar.YEAR`	1	the first year in each era. It should be returned when a long style (`Calendar.LONG_FORMAT` or `Calendar.LONG_STANDALONE`) is specified. See also the Year representation in `SimpleDateFormat`.
`"roc"`	`Calendar.ERA`	0	Before R.O.C.
		1	R.O.C.
`"islamic"`	`Calendar.ERA`	0	Before AH
		1	Anno Hijrah (AH)

Calendar field value names for `"gregory"` must be consistent with the date-time symbols provided by `DateFormatSymbolsProvider`.

Time zone names are supported by `TimeZoneNameProvider`.

Since:

`1.8`

See Also:

`CalendarDataProvider`, `Locale.getUnicodeLocaleType(String)`

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	`CalendarNameProvider()`	Sole constructor.

Method Summary

Abstract Methods

Modifier and Type	Method	Description
abstract `String`	`getDisplayName(String calendarType, int field, int value, int style, Locale locale)`	Returns the string representation (display name) of the calendar `field` `value` in the given `style` and `locale`.
abstract `Map<String,Integer>`	`getDisplayNames(String calendarType, int field, int style, Locale locale)`	Returns a `Map` containing all string representations (display names) of the `Calendar` `field` in the given `style` and `locale` and their corresponding field values.

Methods inherited from class java.util.spi.LocaleServiceProvider

`getAvailableLocales`, `isSupportedLocale`

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

CalendarNameProvider

```
protected CalendarNameProvider()
```

Sole constructor. (For invocation by subclass constructors, typically implicit.)

Method Detail

getDisplayName

```
public abstract String getDisplayName(String calendarType,
                                      int field,
                                      int value,
                                      int style,
                                      Locale locale)
```

Returns the string representation (display name) of the calendar `field` `value` in the given `style` and `locale`. If no string representation is applicable, `null` is returned.

`field` is a `Calendar` field index, such as `Calendar.MONTH`. The time zone fields, `Calendar.ZONE_OFFSET` and `Calendar.DST_OFFSET`, are *not* supported by this method. `null` must be returned if any time zone fields are specified.

`value` is the numeric representation of the `field` value. For example, if `field` is `Calendar.DAY_OF_WEEK`, the valid values are `Calendar.SUNDAY` to `Calendar.SATURDAY` (inclusive).

`style` gives the style of the string representation. It is one of `Calendar.SHORT_FORMAT` (`SHORT`), `Calendar.SHORT_STANDALONE`, `Calendar.LONG_FORMAT` (`LONG`), `Calendar.LONG_STANDALONE`, `Calendar.NARROW_FORMAT`, or `Calendar.NARROW_STANDALONE`.

For example, the following call will return `"Sunday"`.

```
getDisplayName("gregory", Calendar.DAY_OF_WEEK, Calendar.SUNDAY,
               Calendar.LONG_STANDALONE, Locale.ENGLISH);
```

Parameters:

calendarType - the calendar type. (Any calendar type given by locale is ignored.)

field - the Calendar field index, such as Calendar.DAY_OF_WEEK

value - the value of the Calendar field, such as Calendar.MONDAY

style - the string representation style: one of Calendar.SHORT_FORMAT (SHORT), Calendar.SHORT_STANDALONE, Calendar.LONG_FORMAT (LONG), Calendar.LONG_STANDALONE, Calendar.NARROW_FORMAT, or Calendar.NARROW_STANDALONE

locale - the desired locale

Returns:

the string representation of the field value, or null if the string representation is not applicable or the given calendar type is unknown

Throws:

IllegalArgumentException - if field or style is invalid

NullPointerException - if locale is null

See Also:

TimeZoneNameProvider, Calendar.get(int), Calendar.getDisplayName(int, int, Locale)

getDisplayNames

```
public abstract Map<String,Integer> getDisplayNames(String calendarType,
                                                    int field,
                                                    int style,
                                                    Locale locale)
```

Returns a Map containing all string representations (display names) of the Calendar field in the given style and locale and their corresponding field values.

field is a Calendar field index, such as Calendar.MONTH. The time zone fields, Calendar.ZONE_OFFSET and Calendar.DST_OFFSET, are *not* supported by this method. null must be returned if any time zone fields are specified.

style gives the style of the string representation. It must be one of Calendar.ALL_STYLES, Calendar.SHORT_FORMAT (SHORT), Calendar.SHORT_STANDALONE, Calendar.LONG_FORMAT (LONG), Calendar.LONG_STANDALONE, Calendar.NARROW_FORMAT, or Calendar.NARROW_STANDALONE. Note that narrow names may not be unique due to use of single characters, such as "S" for Sunday and Saturday, and that no narrow names are included in that case.

For example, the following call will return a Map containing "January" to Calendar.JANUARY, "Jan" to Calendar.JANUARY, "February" to Calendar.FEBRUARY, "Feb" to Calendar.FEBRUARY, and so on.

```
getDisplayNames("gregory", Calendar.MONTH, Calendar.ALL_STYLES, Locale.ENGLISH);
```

Parameters:

calendarType - the calendar type. (Any calendar type given by locale is ignored.)

field - the calendar field for which the display names are returned

style - the style applied to the display names; one of Calendar.ALL_STYLES, Calendar.SHORT_FORMAT (SHORT), Calendar.SHORT_STANDALONE, Calendar.LONG_FORMAT (LONG), Calendar.LONG_STANDALONE, Calendar.NARROW_FORMAT, or Calendar.NARROW_STANDALONE

locale - the desired locale

Returns:

a Map containing all display names of field in style and locale and their field values, or null if no display names are defined for field

Throws:

NullPointerException - if locale is null

See Also:

Calendar.getDisplayNames(int, int, Locale)

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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